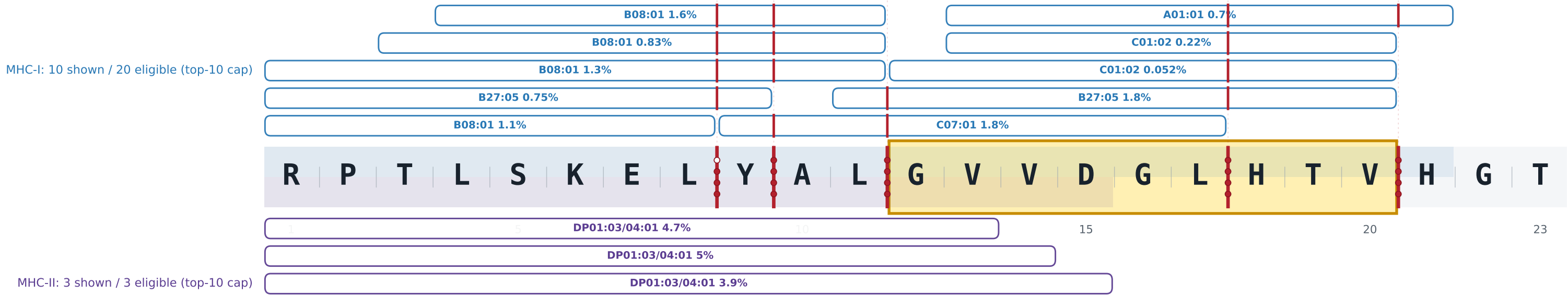
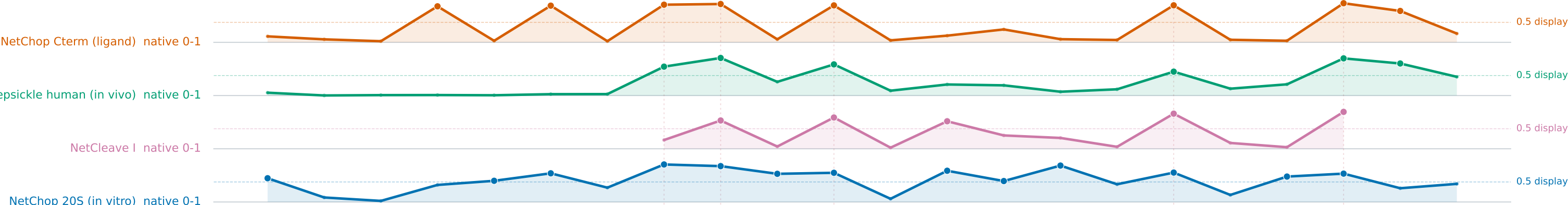


BTD p.Phe445Val | JLF V3, JLF V2, JLF V1 | 23 aa

disclosed minimal epitope GVVDGLHTV at 12-20

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

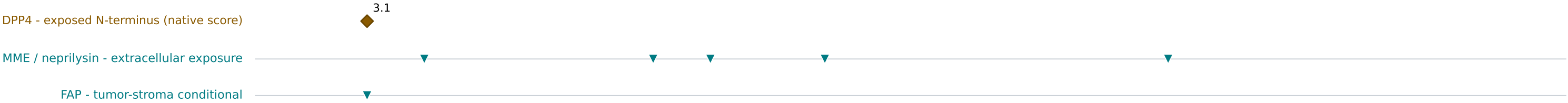


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR BEFORE/DURING SLP UPTAKE - exposure-dependent evidence



SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.